

Effect of information load and cognitive style on cognitive load of visualized dashboards for construction-related activities

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ABSTRACT

A well-designed visualized dashboard could provide intuitive information to construction project managers for effective decision-making. However, the impact of information load of dashboards on cognitive load has rarely been investigated. The roles of user cognitive styles were also ignored. This study examined the effect of dashboard information load and user cognitive style on cognitive load when using dashboards in the construction industry. We designed a visual search experiment with five information load levels. Fifty-six participants, divided into field-dependent (FD) and field-independent (FI) groups, participated in the experiment. The results showed that the FI group had a lower cognitive load and was more efficient than the FD group. The effect of information load on cognitive load followed a piecewise linear correlation with the two prominent knots. The findings underscore the importance of assessing the information load and providing psychological underpinnings for cognitive-fit dashboard design for smart construction management.

1. Introduction

The construction industry is undergoing a smart-pervasive revolution, typified by breakthrough technologies such as the Internet of Things and sensor or vision-based tracking systems embedded on construction sites [1]. The assistance that these technologies bring to construction has resulted in an exponential growth of data from jobsites, which can be converted into valuable information for making educated decisions [2]. Simultaneously, demand for digital and intelligent visualization of construction project information has grown since the Chinese Ministry of Housing and Urban-Rural Development promoted the development of smart construction site-related equipment in 2020. In this context, visualized dashboards play an important role in information presentation during the building of future smart construction sites.

Dashboards are increasingly being used in construction projects, as they are a data-driven site decision support tool capable of conveying critical information regarding overall project performance in an intuitive and clear display format [3]. Various project indicators are often visible at a glance on a dashboard [4]. However, how to efficiently present project information on these visualized dashboards for smart construction sites during their design is not trivial. In terms of information visualization, current dashboards in the construction sector still

have an Achilles' heel. On the one hand, developers tend to use modular design to create dashboard interfaces with much attention to appealing presentation effects and diverse project information. Excessive information gathering under these designs may subject construction project managers to information overload, putting them under tremendous cognitive load, which often distracts rather than concentrates their attention [5]. When confronted with such dashboard displays, like dynamic spatiotemporal but not static low-resolution data in construction [6], they are more prone to make more mistakes by overlooking the most critical information and to waste time making decisions [7]. As a result, an evaluation of how the information presented on visualized dashboards affects users' cognitive load is essential for dashboard design optimization in the construction industry.

Another limitation of the existing dashboards in this industry is that they are typically designed solely based on construction project characteristics, while ignoring individual differences in perceiving information which is linked to a person's cognitive system. A one-size-fits-all dashboard for all construction project managers would be ideal, but it is utopic. Because of individual differences [8], not every user has the same visualization literacy or makes management decisions using the same cognitive brain-based mechanism. Nonetheless, most dashboards used in smart construction management today fall short of being user-

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centric and cognitive-fit designs from a human perspective. It is still unclear how to incorporate individual differences in cognitive styles of management decision-makers, into the development of visualized dashboards in the construction industry.

The current literature on dashboards for construction projects mostly concentrated on the development and optimization of dashboards for construction management with the goal of enhancing their accuracy in practice [4,9,10]. Some other studies highlighted the practical implementation of dashboards like performance assessment on construction projects using these digitally visible tools [11,12]. From a fundamental cognitive perspective of influence mechanisms, there is still a paucity of research on how information load given on dashboards for smart construction management may affect cognitive load of users. Even though recent studies on human-computer interaction (HCI) have looked into how dashboards affect users' cognitive load in different domains, these studies have typically focused on the visual presentation effects of dashboard designs built on visualization forms [13–16]. Few of them are typically concerned with the effect of information given on dashboards in terms of information load. Besides, a construction project requires a considerable amount of high-dimensional information for visualization, like human, machine, material, method, and environment. Construction sites are likewise fraught with uncertainties and complexities, necessitating more regular monitoring of visual information than in other HCI fields. Therefore, research into how dashboards affect cognitive load from an information load standpoint is critical in the construction industry. Additionally, the concurrent effect of users' innate cognitive styles in dashboard development is still overlooked in existing HCI-related studies.

To bridge the above knowledge gaps, this research aims to investigate the impact of dashboard information load and user cognitive style on cognitive load for construction-related activities. The ensuing sections begin with existing work related to the development and user impact of dashboards in HCI areas, particularly in the construction industry. Thereafter, research hypotheses will be proposed. In the fourth section, the whole methodological framework of this study will be illustrated in detail. The results of the cognitive style test, quantification of information load, task performance, and eye movement will be presented in the fifth section, followed by a discussion of the corresponding findings. The final section concludes the paper.

2. Review

2.1. Dashboard development in HCI

A dashboard is a visual depiction of key information from numerous sources that helps users make informed decisions in real time by combining and analyzing a wide range of data [17]. How to develop and design a visualized dashboard has become a hotspot in many domains, like smart city dashboard for city management and life quality improvement [6], healthcare dashboard for patient records [18], business intelligence dashboard for business analytics [19], automobile dashboard for traffic alerts [20]. In these studies, the dashboard concept has been evolved and can serve an expanding variety of various purposes, such as monitoring, planning, communication, learning, and motivation [21]. Dashboards are frequently classified according to the kind of decision-making they assist, including operational for front-line practitioners and strategic or tactical for upper management [22]. As for the forms of visualized dashboards, they can be divided into two display protocols of head-up displays and head-down displays depending on the modality [23]. With the emergence of virtual and augmented reality (VR/AR), dashboards can now be classified as further subsets of technology-dependent interactive interfaces, such as hybrid user interfaces that mix cross-reality devices (e.g., AR head-mounted displays) with smartphones [24,25], in addition to traditional screen-based monitoring displays.

While research on visualized dashboards exists in many fields, there

are relatively fewer studies in the construction industry, providing limited insight for practitioners and scholars. In terms of dashboard types in this sector, interest is developing from the static visual genre to the functional genre [21], which allows real-time monitoring of the representative status of construction projects and supports user intervention based on dynamically updated data in response to the status [11,26]. Some early researchers also began to concentrate on interactive interfaces and developed an augmented reality visualization for construction work based on building information modeling [27]. Owing to the powerful utility for information readability and comprehensibility, dashboards have been frequently developed to assist construction project managers with worker identity verification [28], early warning and safety monitoring [9], progress tracking and reporting [4], portfolio management [3], project performance analysis [11], etc. Therefore, the application of dashboards can achieve many purposes in construction practice, including better data integration and delivery, more effective project decision-making, cost savings from eliminating manual reporting, use as a data warehouse to serve as a reference for future project practices, improved stakeholder coordination, and enhanced worker wellbeing [4,11,29]. When it comes to dashboard users, project managers are the group that uses dashboards for decision-making the most frequently in the construction industry, but dashboards can be designed to fulfill a number of goals to target different users, from the top government to frontline engineers [26,30]. In general, studies on construction-related dashboards majorly focused on better technological design and development of dashboards from a project-centered perspective. Others highlighted how the use of accessible dashboards might help construction projects achieve their various goals more efficiently. None of them, however, considered how these smart visual interfaces would influence users' cognitive load from a user-based viewpoint, leaving the construction industry without any cognitive-fit designs for visualized dashboards.

Although the dashboards in the construction industry are in flux, monitoring displays remain the mainstream and project managers are the most common user based on the above review. Therefore, according to the concept of a dashboard for construction projects provided previously [11], it is defined for the purpose of this research as a digital screen-based visualization built on enterprise intelligence and data integration infrastructure that assists project managers in more efficiently monitoring and managing high-dimensional information of construction projects, including man, machine, material, method, and environment, as well as overall project performance.

2.2. Dashboard impact on cognitive load

Cognitive load refers to the mental workload imposed on a user caused by cognitive demands in the context of the theory of limited cognitive resources [31], limited span of attention, and limited capacity of working memory. The cognitive load theory has stated three types of cognitive load: intrinsic load, extraneous load, germane load [32]. Intrinsic load is the inherent difficulty of a task and is not subject to artificial modification, whereas extraneous load is influenced by the way data is interpreted and tasks are presented. Effort during learning can cause germane load. Although a visual board was intended to match cognitive load, inappropriate exhibitions on it may have the opposite effect and overburden cognitive load [33]. Extraneous load is the type of cognitive load that we focused on in this research for research aims, because it can be affected by visualization commits [34]. Besides, retrieving information that is irrelevant to learning increases the number of interacting elements in working memory and may result in an extraneous cognitive load that can be detrimental to learning [35].

Early studies on the impact of visualization on user cognition concentrated on comparing basic graphs and tables [36], while researchers are now discussing the cognitive load caused by HCI-related dashboards with rich data visualization due to computing power facilitation [37]. Relevant research on how dashboards affect cognitive load

of users mainly concerned about the visual presentation effect of dashboard design, including visualization layouts [37], color combinations [13], information clarity states [14], presentation speed [15], and graph properties [16]. However, what they highlighted is how cognitive load is affected by visualization forms or styles of dashboards, but not the informational content provided on dashboards in a view of information load. Excessive information on the dashboard is regarded as the most critical issue that produces a cognitive load that overwhelms a user [7], but there is little agreement on the concept of information load in the HCI literature. Several management-related studies took its definition for granted and focused solely on information overload [38], which means brain performance must suffer even if extra information is cognitively processed [39]. According to Errol's research [40], information load was simply information quantity, regardless of repeated or different dimensions. The appropriate range of dashboard information load including different information qualities is still not adequately defined or quantified in HCI research.

Previous studies related to information systems have suggested that the increasing amounts of information exerted varying effect on user psychology, such as on decision-making efficiency [41], information usage judgment [42], and satisfaction with jobs and communication [43]. Nonetheless, these research objects were not directly tied to cognitive load, and thus were unable to show how much mental effort a user needed in a HCI task. Furthermore, compared to the fields of automobile, manufacturing, clinic, business, or e-learning, a construction project is more likely to be a cross-disciplinary, whole-process, collaborative effort that includes numerous components of information for visualization [44]. This needs more cognitive resources while monitoring visual information on construction sites than in other types of HCI workplaces. However, there has been little investigation into the impact of dashboard information load on cognitive load of users in earlier construction-related HCI research in an objective way. The principles behind visual cognition that how dashboard information load affects cognitive load in the construction industry remain largely unclear.

2.3. Cognitive factors influencing information processing

Individual differences were seen as another important type of factors influencing HCI or information perception efficiency in working contexts [45], in addition to information presentation properties on a dashboard [46]. Early evidence has suggested that the quality of a visualization-based judgment may be more user-dependent than format-dependent [47]. Afterwards, more HCI research began to provide fresh insights into the effect of individual characteristics on user cognition in information processing [36]. Specifically, information perception preferences can be affected by many personal matters, such as motivation and interests, educational background, domain expertise, personality, working memory, cognitive style, spatial ability, and demographic factors [48]. As a result, a user-based design of visualized dashboards should consider individual differences for various types of users [49].

From a personal standpoint, cognitive capacity and cognitive style are two major cognitive factors that determine how a person processes information. Nonetheless, with structure preview design, the association between the former one and user performance seems no longer significant [50]. Although cognitive capabilities are critical for information processing, prior research suggests that cognitive style is a better predictor of a user's choices in a given situation than cognitive ability [51]. Besides, some studies agree that affective and intentional factors are less important than cognitive factors [48]. Cognitive style refers to the preferred modes or relatively consistent manner in which an individual acquires, organizes, processes, and interprets information [52], which directly influences the strategies for perceiving, thinking, selecting, problem-solving, and decision-making [53]. The mismatch between visual presentation and cognitive style might lead to lower problem solving and increased cognitive load of users according to cognitive-fit

theory [54]. Although psychology research directed more attention to the moderating effects of user personal characteristics, such as domain expertise [55], and cognitive capacities like numeracy or literacy [56,57], it is still unclear whether cognitive style influences user cognitive load in the visual cognition mechanism when it comes to HCI dashboard design.

Cognitive styles could be subdivided into different dimensions, such as field-dependent (FD) or field-independent (FI), global or local, verbal or imagery, internal or external, and liberal or conservative [58]. Among these categories, FD and FI individuals are closely related to perceptual abilities and visual preference patterns [59], making them more ideal than other groups for visualized dashboard research. Besides, the extraneous cognitive load is assumed to be directly proportional to the degree of intrusion caused by irrelevant information [60], and the difference between these two styles is a clear manifestation of anti-interference ability for irrelevant information. This can appropriately aid to give a thorough insight of user differences in information processing when users face a displayed dashboard with high-dimensional information from construction projects. Therefore, we chose the division of FD and FI as sub cognitive style categories in this research.

2.4. Summary for knowledge gaps

Overall, present studies on construction-related dashboards majorly focused on the technological design and development of dashboards from a project-centered perspective [4,9–12] rather than a cognitive-fit viewpoint that takes user differences into account. It is yet unclear how these visual interfaces in smart construction management would influence users' cognitive load. Most HCI research in other domains is concerned with how visualization forms or styles of dashboards influence cognitive load [13–16], but not the content provided on dashboards in terms of information load. They also failed to consider whether users' intrinsic cognitive styles affect cognitive load when different user types are presented with dashboards. While in construction practices, multiple dimensional information is necessary to present on dashboards [44], which may interfere with project managers making intelligent and efficient decisions, and they may have various cognitive styles owing to personal knowledge or past experiences. Therefore, to develop better and more appropriate dashboards for smart construction management, it is necessary to thoroughly investigate how varying levels of dashboard information load affect the cognitive load of visualized dashboards depending on different users' cognitive types in the construction industry.

3. Hypotheses

An interface containing a low amount of information may not exert a too strong effect on cognitive effort [61]. Users are more inclined to focus on private thoughts and their attention is more likely to wander to task-irrelevant objects when low visualization is present [62], which is counterproductive to dashboard design. The fixation will become longer if an object is challenging to process than if it is simple to process [63]. Hence, users need to devote significant attention and time to examine and comprehend a display with various and numerous information. Under the condition of sufficient cognitive resources that can be available, an increasing amount of information can make users focus and generate enough cognitive resources to deal with the acceptable information load caused by an interface itself [61,64]. However, limited cognitive resources of users are likely to be robbed and overwhelmed by excessive information when an interface becomes quite complex. Based on the attention load theory, a person may fail to have enough available cognitive resources for stimuli processing when he suffers from very high levels of perceptual load. This will result in a low level of mental effort for perform visual tasks, with fewer fixations and less time due to worry, annoyance, stress, and confusion [65]. Therefore, we speculate that a visualized dashboard with moderate levels of information load

which can match users' cognitive load well can be more attractive to users when they apply them for smart construction management. Although there is no evidence directly shown how information load of dashboards affects cognitive load in the existing literature, previous studies have indicated that the effect of information load on cognitive responses [66] and decision quality [67], follows an inverted U-shaped curve. Additionally, some stated that there is an inverted U-shape impact pattern on cognitive load under the influence of website complexity [61], learning material disfluency [68], and extrinsic stress [69] even though the cognitive load in these studies was no longer divided into finer-grained kinds. Similarly, we hypothesize that there is the existence of the following relationship between information load and cognitive load of dashboards:

H1. The relationship between information load and cognitive load takes the form of an inverted U shape, with cognitive load increasing with growing information load but decreasing when information load becomes too excessive on a dashboard.

Cognitive styles are long-lasting psychological characteristics that represent an individual's distinctive manner of processing information [70]. FD and FI styles, which reflect the personal features of detecting details in a complex object, are widely known and have been the most studied over the years. These two groups were found to have distinct visual approaches with FDs being more holistic and FIs more analytical [71], and they were also considered to influence personal preferences during the information processing process [58]. Scholars focused primarily on multimedia learners when interacting with a hypermedia environment in terms of the impact of cognitive styles (FD and FI) on cognitive load. Prior study evidence from questionnaire investigation revealed that FD students always had a higher level of total cognitive load than their FI counterparts, regardless of whether they utilized software with or without metacognitive scaffolding [72]. In another eye tracking-based research, FD learners had a marginally higher cognitive load value than FI learners, while having the same degree of cognitive load [35]. Nonetheless, little is known about the role of such cognitive style difference in cognitive load during information processing when users are presented with a visualized dashboard related to smart

construction. Hence, the following hypothesis is proposed based on the alleged stylistic difference:

H2. The cognitive load of FD users is higher than that of FI users with the same information load.

4. Methodology

The information load can be altered by altering the level of information in demand or the available mental resources committed to a specific task [38]. This principle guided our research and we followed earlier similar studies [66,67], which gradually increased the number of attributes from a low level of information to a high level. Therefore, a laboratory experiment was carried out with a 5×2 design (i.e., 5 levels of dashboard information load (between-subject) $\times$ 2 types of cognitive styles (within-subject)). A comparative observational research design was utilized to evaluate the effect of cognitive styles (i.e., FD or FI) on participants' cognitive load when confronted with dashboards with varying information load levels. The research workflow, comprising experimental implementation and data analysis in this study, is shown in Fig. 1.

4.1. Experiment

4.1.1. Experimental apparatus

Eye tracking was adopted in this research to measure cognitive load because the dashboard is a visualized tool depending mainly on visual perception, and eye movement has high-efficient and precise performance in visual attention allocation and visual computing in HCI research [73]. The activity of eyes, as the major doorway to the brain, can objectively expose cognitive processes and provide insights into mental processes and intents [73]. In addition, compared to other physiological metrics (e.g., electroencephalography, galvanic skin response, and heart rate variability), eye tracking is a non-invasive approach for measuring cognitive load in real time [74], which can be used better and more feasibly when users utilize dashboards in construction practices.

The eye-tracking equipment used in this experiment was a Tobii Pro

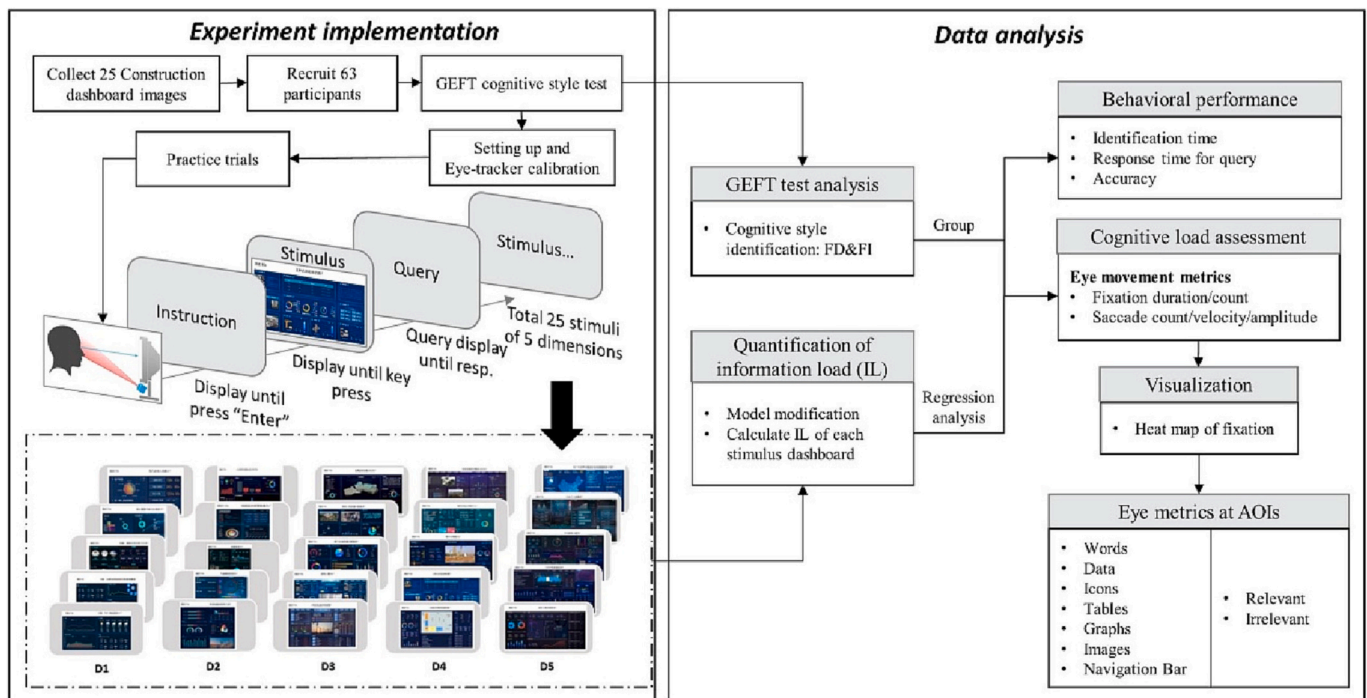


Fig. 1. Experiment and data analysis workflow.

Fusion eye tracker (Manufacturer: Tobii AB, Trademark: Tobii Pro, Model: Fusion) mounted on a 24-in. desktop with a computer screen resolution of 1920×1080 pixels (Fig. 2). The eye tracker has a sampling rate of 250 Hz, so it can precisely monitor a wide range of eye positions and gazes at any moment, which is particularly useful for saccade-based research in terms of time or areas of interest. The eye tracker performs a single-target calibration procedure. Tobii locates a specific target in the scene image and corrects for the eye position biases and drifts in response to an eye calibration request. It gives display-centered coordinates after calibration is complete. The embedded stereoscopic technology and dual-camera system provide superior tracking precision and robustness for head-movement tolerance. The manufacturer claims the nominal accuracy of the eye tracking reaches 0.6° . The Tobii Eye-tracker Manager and Analytics Pro-lab Software [75] were responsible for handling stimuli presentation and recording data.

4.1.2. Participants

63 participants (undergraduate and postgraduate students) majored in civil engineering or construction management (34 women). Seven participants were removed due to unsatisfactory eye calibration, leaving 56 participants (30 women) as the final samples. The participants were all right-handed and had limited construction experience, ranging in age from 18 to 30 years. All participants were deemed acceptable for the eye-tracking test because they had no vision impairment or eye illnesses. The Group-Embedded Figure test (GEFT) proposed by Witkin [76] was used to classify the participants into two groups: FD and FI. FD individuals are thought to have more difficulty isolating information from their context and are more affected by surrounding signals than field-independent individuals. FI individuals tend to identify objects or details by separating elements from a complicated background [58]. Participants were classified as FD or FI based on the median of the total test score as the cutoff point according to previous study evidence [77,78], with the FD score being lower than the median and the FI score being higher than the median. This study was conducted in accordance with ethical guidelines and approved by the Ethical Committee of the City University of Hong Kong.

4.1.3. Dashboard stimuli

Five levels of dashboard information load were designed, which were greater than three levels to exhibit the increase and decrease in information processing, a necessary design condition to indicate that overload may occur [66]. The number of information attributes was considered to be a crucial element to manipulate different information load levels [66,67]. Nonetheless, it may not be suitable for manipulate dashboard information load in the construction industry. This is because a construction project is characterized as multidimensional information and a dashboard for smart construction management involves too many information cues compared with any other daily visualization. Given that modular designs are typically employed in this sector, we used the number of information modules rather than information attributes to

manipulate dashboard information load levels. The ideal number of information characteristics has been estimated to be between 4 and 15 [79]. The final number of separate information modules on each level in this experiment was decided as 4, 6, 9, 12, and 16, respectively (as shown in Fig. 3). This also corresponds to the popular number for actual modular design on construction dashboards. Each level of information load stimuli comprised five dashboards used in practice for smart construction management, which were all collected from the Baidu search engine. These dashboards may involve the following items according to project differences: employee monitoring (employees' working hours, information of employees by types of work, entry records, etc.) and environmental monitoring (Air Quality Index, PM2.5, etc.), equipment operation monitoring (operating parameters, operation time of machines, etc.), the status of the project (project stakeholders, project progress, etc.), intelligent video surveillance, safety inspection (disaggregated statistics of safety hazards, types of inspection, etc.), energy consumption monitoring (water and electricity consumption), early warning information (alarm events, number of alarms), material monitoring, and quality monitoring.

4.1.4. Task design and procedure

Cognition is the process of learning and understanding that is typically associated with the question of "what" (e.g., what happened, what is going on, what does that knowledge mean) [48]. In order for participants to complete a cognitively demanding visual search task, a what-related question (e.g., what is the noise level shown on the current dashboard, which day of the week has the fewest workers on the job, what is the pass rate for decorating projects) was correspondingly created based on context items on each dashboard. The question on each dashboard was standardized to be in the same upper location. Each question had four possible answer responses: one correct response, two obfuscated incorrect responses, and a fourth "Not found" response. These choices appeared on the query display that appeared after each dashboard display. Therefore, a total of 25 dashboard displays were presented in a row, each followed by an query display. Participants were required to figure out the answer to the question on each dashboard display as soon as possible, and were then asked to choose their response on the following query display. The current display would not transition onto the subsequent display until the participant pressed a "Enter" button on the current display. A pilot test was conducted before the formal experiment was carried out, and adjustments were made to the time constraints. The experimental task and protocol were firstly introduced to the participants, and they willingly signed an informed consent form after careful consideration. The GEFT was subsequently administered to students to assess which cognitive type they fit into. Participants were given adequate practice time to ensure that they thoroughly grasped the entire process. Dashboard stimuli used during the practice stage differed from those used during the formal experimental stage. At the start of the formal stimulus series, each participant's position (approximately 70 cm away from the 24-in. display) and a calibration procedure using a nine-point setup were completed. Each blinking point emerged at the center and in the four inner or outer corners of the screen. Participants were instructed to focus on the blinking point until both eyes had satisfactory mapping accuracy and precision ($<1^\circ$ visual angle). They were all given the guidelines following calibration, and were right to start trialing once they were prepared.

4.2. Data analysis

4.2.1. Identification of cognitive styles

Participants were required to identify whether the presented simple figure was embedded in the complicated figure displayed for each item, as shown in Fig. 4. A Chinese version of the GEFT revised by the College of Psychology in Beijing Normal University was employed [80], which is a mature perceptual test used in the current literature [81]. The test

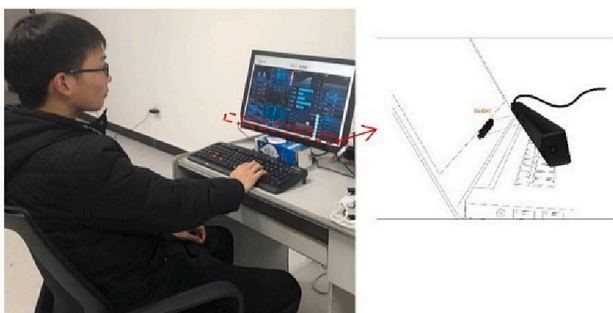


Fig. 2. The experiment scene consists of a Tobii Fusion eye tracker and stimuli presented on screen.



Fig. 3. Dashboard examples based on information module numbers (D1 with 4 information modules, D2 with 6 information modules, D3 with 9 information modules, D4 with 12 information modules, D5 with 16 information modules).

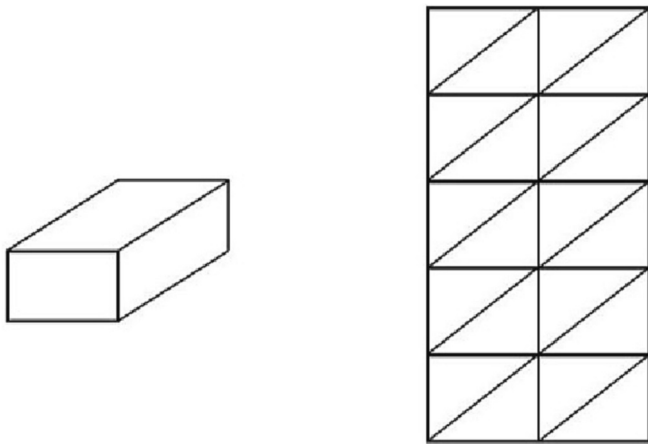


Fig. 4. Example from the Embedded Figure Test: the left is a simple figure, and the right is an embedded figure.

consists of 25 items divided into three sections, with the first section (7 items) having a time constraint of 2 min and intended to familiarize and practice participants while disregarding score computations. The answers to the second (9 items) and third (9 items) sections provide score computations with one point for each item. The total score for the test items was 18. Those who scored lower than the median score of 18 were labeled as FD, and those who scored above the median score were assigned to FI. Individual cognitive styles are enduring and stable personality traits that are formed based on not only personal but also social factors, such as preferences, beliefs, childhood experiences, or learning environments [51,82], but will not change with the specific time or context changes. Meanwhile, in order to prevent the practice effect, the test was therefore given once to each participant, as in prior related investigations [35,70].

4.2.2. Quantification of information load

A dashboard has a great number of information cues and design elements. Therefore, despite the fact that the number of information modules increased over the levels, this research measured dashboard

information load using a set of scoring criteria based on the actual visualization characteristics that influence dashboard information load other than information quantity. We computed the information load using the study of Thomas and Pourya [83], which developed a theory-based model that can be quantified for estimating information overload from a systematic view. Seven factors covering information, individual, and task, were considered, as these three are the fundamental building blocks that might lead to information overload in the individual landscape. According to Dubin's categoric type of law, they are separated into two factor categories: "Against" and "For" factors [84]. Specifically, information characteristics (IC), information processing capacity (IPC), and available time (T) are "Against" factors, and their collaboration can lessen information load. In contrast, "For" factors include personal factors (PF), task and process parameters (TPP), information quality (Qual), and information quantity (Quant), and they might interact to enhance the chance of information overload. In accordance with Dubin's law, the emergence of information overload can be metaphorically described as unbalanced, depending on two sides of a scale with "Against" factors on one side and "For" factors on the other. Individuals will become overburdened by a massive amount of information if the weighting of "For" factors exceeds that of "Against" factors. Individuals, on the other hand, can cope with information load within their cognitive limits if the weighting of "For" factors is smaller than that of "Against" factors. Notably, more sophisticated information may be considered greater quality, but also requires more time to process. This can be compensated for by multiplying information characteristics by both information processing capacity and available time in terms of the interactions among "Against" factors, yielding a "not-overloaded" score as opposed to a combined "overloaded" score [83]. The consideration for factor interaction within the "For" factor category is comparable. The calculation makes the assumption that the overall weighting of two factor categories is the same. As a result, based on the metaphorical relationships of these two factor categories, information load can be calculated by their subtraction with using multiplication to reflect factor cooperation within each factor category [83], as shown in Formula (1):

$$IL = (IC \times IPC \times T) - (PF \times TPP \times Qual \times Quant) \quad (1)$$

where IC denotes information characteristics (a weighting range of 0–10), IPC denotes information processing capacity (a weighting range of 0–100), T denotes available time (a weighting range of 0–100), PF

denotes personal factors (a weighting range of 0–10), TPP denotes task and process parameters (a weighting range of 0–10), Qual denotes information quality (a weighting range of 0–10), and Quant denotes information quantity (a weighting range of 0–100). The weighting variation of each factor is consistent with that provided in the original theory-based model, indicating the importance of one factor over another [83]. Hence, the information load value of a dashboard could fall anywhere between –100,000 and 100,000. When the formula result returns a value of zero, it indicates a transition or tipping point between the unloaded and overloaded states. A result of less than zero indicates that the individual is overwhelmed by information, whereas a result greater than zero indicates that the person is not overburdened with information.

The components of IC in Formula (1) include complexity, ambiguity, and uncertainty [85], which were calculated using Formula (2). Based on the features of the collected dashboard features and the relative relevance of components, the complexity range was assumed from 0 to 20, the ambiguity range from 0 to 2, the uncertainty range from 0 to 0.5, and the novelty range from 0 to 0.5. It claims that the lower the score, the more complicated, ambiguous, uncertain, and innovative the information.

$$IC = \text{complexity} \times \text{ambiguity} \times \text{uncertainty} \times \text{novelty} \quad (2)$$

According to the parameters of the experiment in this study, complexity was separated into five parts: question complexity, response complexity, answer font size, the distance between answers and questions, as well as layout regularity. They were all supposed to have the same highest possible score of 4 on average according to the total weighting of complexity. As mentioned, the formula was constructed using the same basic logic as the entire computation procedure. Specifically, query complexity was primarily assessed with considering whether the question is simple to comprehend based on the length of the given question. The length of questions in this experiment spans from 8 to 28 words, which were then normalized to 0–1. The query complexity was then scored on four levels based on quartiles of the normalized range. The higher the normalized word count, the more complex the question, and thus the lower the score. Response complexity was graded by professionals on a scale of 1 to 4, with higher complexity resulting in a lower score. This assessment was made based on how easily the answer could be found on the interface, with comprehensively considering whether the answer could be directly located (e.g., direct: “how many workers entering the site on Monday?” vs. indirect: “Which day of the week has the least number of workers entering the site?”), whether the dashboard contains the same key words in the question for obtaining answers, and so on. The assessment of answer font size and distance on dashboards was similar to that of query complexity, based on the actual font size and distance ranges. Furthermore, the information modules in the acquired dashboards were displayed in either an organized matrix format or a non-matrix format, with the latter being considered more complicated. They were graded by professionals on four levels, with less regular layout resulting in a lower score.

$$\text{Complexity} = \text{query complexity} + \text{response complexity} + \text{answer font size} + \text{distance} + \text{layout regularity} \quad (3)$$

Compared with other components in Formula (1), IPC and T were thought to be single-factors in nature. The higher the IPC score, the better the capacity to process information. IPC was fixed at 50 in this experiment due to the fact that participants were all students with limited previous construction expertise based on an arbitrary weighting rule. The lower the score for the time evaluation, the more urgent the situation; the higher the time score, the more time was available. Due to the task requirement for participants to locate answers as soon as possible, T for task completion was set to a relatively low score of 30, indicating that cognitive load would be induced under time-critical conditions.

Personal factors are thought to contribute significantly to information overload [41]. Notably, personal factors here refer to a form of extraneous factor that indirectly affects information load and consists of numerous components combining the level of prior experiences, personal skills, cognitive style, motivation, and personal situation, seen in Formula (4). It reflects intrinsic and long-established personality traits that influence the amount of information individuals receive as well as the strategy or coping methods they use during the individual process for completing interactions and internal computations. In other words, it has nothing to do with one's current cognitive states in a specific context. To account for these various individual characteristics mentioned above and meanwhile to simplify the calculation, the formula employs the proposed quantification approach in the developed theory-based model [83]. It states that each component can have a maximum score of 2, with a higher aggregate score indicating a negative personal impact. Similarly, given the participants' homogeneity, they all received the same score on personal factors based on their actual individual characteristics: the experience aspect was scored 2 with considering a lack of occupational experience on construction sites; the skill component was scored 1 because they have ability to interpret words and figures; the cognitive style component received a score of 2 for taking into account style-based impact variations; motivation was given a score of 1 due to the fact that they would be inspired by experimental demands despite a lack of actual motivation for site monitoring, and situation was scored 1 for the assessment of their relevant professional background.

$$\text{Personal Factors} = \text{level of prior experiences} + \text{personal skills} + \text{cognitive style} + \text{motivation} + \text{personal situation} \quad (4)$$

TPP consists of task complexity, task novelty, task interruption count, and multitask count, as illustrated in Formula (5). Greater task difficulty resulted in a higher score being assigned to this component on the basis of information overload calculation. Because no interruption or multitasking settings were utilized in this experiment, the maximum weighting of task complexity and novelty was adjusted to the average of the highest weight of TPP as 5. Task complexity increases as query complexity and search difficulty increases. As a result, dashboards with the same number of information modules were presumed to have the same level of task difficulty, and each level scored one point higher than the lower level, with a starting score of 2 assigned to dashboards with only four information modules.

$$\text{Task and process parameters} = \text{task complexity} + \text{task novelty} + \text{number of task interruption} + \text{number of other task(multitasking)} \quad (5)$$

Validity (0–1) and relevancy (0–10) were used to rate information quality, with a higher number indicating strong validity and high relevancy. It can be quantified, according to Slawson et al. [86], by multiplying validity by relevancy, which both have a compound impact on information quality. Because all of the dashboards were obtained from web engines and have already been utilized in smart construction practice, the validity and relevancy are deemed convincing, with a score of 0.9 and 9, respectively.

$$\text{Quality of Information} = \text{validity} \times \text{relevancy} \quad (6)$$

With regard to Quant, we partitioned the entire amount of data into relevant (0–50) and irrelevant information (0–50) to the displayed question and subdivided each part into subsections using Formula (7). Each subsection has a maximum weight of 5 according to the same principle of equal weighting distribution, with a higher score signifying a greater volume of information. The subsections cover the most common information formats found on visualized dashboards in the construction sector. Based on their actual numbers on the dashboards, their scoring criteria were similar to that of query complexity. The relevant information contained query words, answer-related data, semantic words, and other assessments located in the answer zone.

$$\begin{aligned}
\text{Quantity of Information} = & \text{the number of words} + \text{the number of data} \\
& + \text{the number of tables} + \text{the number of figures} \\
& + \text{the number of icons} + \text{the number of images} \\
& + \text{the number of information forms} + \text{the number of modules} \\
& + \text{the number of indicators} + \text{the number of variables}
\end{aligned}
\quad (7)$$

4.2.3. Behavioral patterns

To evaluate time spent and response speed, the identification time (IT) for observing each dashboard and the response time (RT) for picking an answer were calculated. Response accuracy (RA) was also measured to illustrate how many accurate responses participants chose based on boolean values, with 0 and 1 representing incorrect and correct responses, respectively. The cognitive style was used as a within-subject variable to examine the influence of group on task performance. Because the dataset violated Levene's test for equality of variance and the Shapiro-Wilk test for normality, the non-parametric Mann-Whitney (two-tailed) test was employed to compare task performance between groups.

4.2.4. Eye movement patterns

Trials with poor calibration (above 1° visual angle) and >30% missing gaze were deemed unreliable and discarded from the study, eventually resulting in seven participants being excluded from further analysis. Tobii I-VT fixation filter was adopted in this paper, the velocity threshold parameter is set to 30°/s, and the speed calculator timespan is set to 20 milliseconds. The average gaze locations of the left and right eyes were used as the analytical units.

The time of interest (TOI) was defined as the presence of each dashboard. Average eye-tracking metrics, including the most used fixation-based and saccade-based metrics within the TOI, were analyzed at the group level to assess cognitive load. We used the Mann-Whitney *U* test to test whether TOI-based fixation metrics varied with cognitive style. Two methodologies were used to classify the area of interest (AOI): functional division and relevance division. Functional division divides the information in the dashboard into seven functional areas: data, tables, graphs, icons, images, navigation bars, and the words of each dashboard. The "graphs" AOI includes title legend and horizontal and vertical axis, and "data" AOI refers to a snapshot view of critical data indicators. The differences between the FD and FI groups were compared using the Mann-Whitney test, whereas the differences between the relevant and irrelevant zone percentages were compared using the paired *t*-test. Finally, a natural spline regression analysis was conducted between cognitive load evaluated by eye movement metrics and information load to determine the changing trend of cognitive load with information load. We performed a linear mixed model (LMM) regression analysis to discover specific information factors and group variables on cognitive load. All statistical and regression analyses were performed using the R Studio software [87].

Metrics for AOIs analysis were fixation-based indicators: average fixation duration, fixation count, dwell percentage of each functional area, and time to the first fixation in the relevant or irrelevant information area which were divided according to the relevance of the content to the displayed question. Table 1 summarizes the metrics used for further analysis.

5. Results

5.1. Group embedded figure test (GEFT)

Regarding cognitive style, GEFT scores varied from 2 to 18 (Table 2). The average score was 9.64 (median = 9.5, SD = 3.39) across valid 56 participants, which was distributed normally according to the Shapiro-Wilk test ($p > 0.05$). According to the median score, participants were separated into two groups: FD and FI. The FD group comprised 28 participants with $2 \leq \text{GEFT} \leq 9$, whereas the FI group consisted of 28

Table 1

TOI-based and AOI-based metrics used in eye-tracking data analysis.

Type	Metric	Indication
Fixation-based	Total fixation duration	Longer fixation duration indicates an increased load on the working memory [88,89]
	Average fixation duration	More fixations reflect higher cognitive load [88]
	Fixation count	More saccades indicate higher cognitive load [90]
Saccade-based	Saccade count	An increase in saccade velocity indicates a higher task difficulty; a decrease in saccade velocity indicates tiredness [91]
	Average velocity of saccades	Longer saccades indicate higher load [92]
	Average amplitude of saccades	
Dwell time-based	Dwell percentage	A typical metric used to assess changes in visual attention allocation for different AOIs [93]
First fixation-based	Time to the first fixation	A typical metric for visual recognition time that is interfered by irrelevant information [94]

participants with $9 < \text{GEFT} \leq 18$. Participants in the FD group had an average GEFT score of 6.89 (SD = 1.89), whereas those in the FI group had an average score of 12.39 (SD = 2.03).

5.2. Information load of dashboard

Table 3 shows the information load of 25 dashboards derived using the formulae in Section 4.2.2. Based on the previously defined scoring criteria, the results of information-processing capacity (IPC), available time (T), personal factors (PF), and information quality (Qual) for all dashboard stimuli are identical. In general, information-related components, such as information characteristics (IC) and information quantity (Quant), differ significantly and play a larger role than others in influencing the information load across all dashboard stimuli. The information load of the gathered dashboards ranges from −20,460 to 10,337.70, with a mean of −2898.02 and a standard deviation of 8900.77, indicating that stimuli are highly heterogeneous. The dashboards featuring nine or more information modules have negative information load values less than zero, indicating that these dashboards may overload participants.

5.3. Task performance

Table 4 presents the task performance according to cognitive styles. All performance indicators differed significantly between the FD and FI groups ($p < 0.05$). The FI group identified the objects and responses with a shorter duration than the FD group ($p < 0.05$). The FI group also had higher response accuracy than the FI group ($p < 0.001$). Task performance for each stimulus at the group level was calculated, as shown in Fig. 5. Although identification time of the FI group was lower than that of the FD group on average, statistically significant differences between the two groups were only observed at the s13 ($\Delta = 12.77$ s, $n = 56$, $p < 0.05$) and s14 dashboards ($\Delta = 7.71$ s, $n = 56$, $p < 0.05$) with the moderate level of information load. Similarly, response time for most stimuli were shorter in the FI group than in the FD group, but the statistical difference between the two groups only appeared on the s22 dashboard ($\Delta = 1.21$ s, $n = 56$, $p < 0.05$). The accuracy of the FI group was higher than that of the FD group, especially when they encountered dashboards with 12 or 16 information modules. Significant differences were observed between the two groups regarding the s21 ($\Delta = 29\%$, $n = 56$, $p < 0.05$) and s22 ($\Delta = 33\%$, $n = 56$, $p < 0.05$) dashboards which both belong to those with 16 modules.

Table 2
Results of GEFT cognitive style test.

Cognitive style	N	Minimum	Maximum	Mean	SD	Lower 95% CL of mean	Upper 95% CL of mean
FI	28	10.00	18.00	12.39	2.03	11.61	13.18
FD	28	2.00	9.00	6.89	1.89	6.16	7.63
Valid N	56	2.00	18.00	9.64	3.39	8.74	10.55

Table 3
Quantification scores of information load of each dashboard stimulus.

Stimulus	IC	IPC	T	PF	TPP	Qual	Quant	IL
1	9.5000	50	30	7	2	8.1	34.5	10,337.70
2	8.5000	50	30	7	2	8.1	36.0	8667.60
3	9.2500	50	30	7	2	8.1	35.5	9849.30
4	7.2000	50	30	7	2	8.1	37.5	6547.50
5	8.5000	50	30	7	2	8.1	36.5	8610.90
6	6.5000	50	30	7	3	8.1	46.5	1840.35
7	6.6500	50	30	7	3	8.1	44.5	2405.55
8	6.9000	50	30	7	3	8.1	43.5	2950.65
9	6.5000	50	30	7	3	8.1	43.5	2350.65
10	6.5500	50	30	7	3	8.1	43.0	2510.70
11	6.1500	50	30	7	4	8.1	50.0	-2115.00
12	6.5000	50	30	7	4	8.1	48.0	-1136.40
13	6.1500	50	30	7	4	8.1	52.0	-2568.60
14	6.7500	50	30	7	4	8.1	50.0	-1215.00
15	5.6250	50	30	7	4	8.1	48.0	-2448.90
16	6.3500	50	30	7	5	8.1	53.5	-5642.25
17	6.7500	50	30	7	5	8.1	61.5	-7310.25
18	6.5000	50	30	7	5	8.1	54.5	-5700.75
19	5.1425	50	30	7	5	8.1	62.5	-10,005.00
20	6.0500	50	30	7	5	8.1	62.0	-8502.00
21	6.2000	50	30	7	6	8.1	65.5	-12,983.10
22	6.2500	50	30	7	6	8.1	70.5	-14,609.10
23	5.7500	50	30	7	6	8.1	71.5	-15,699.30
24	5.0500	50	30	7	6	8.1	75.5	-18,125.10
25	4.0500	50	30	7	6	8.1	78.0	-20,460.60

Table 4
Identification time, response time, and accuracy of the FD and FI group.

Indicator	FD		FI		p-value	Hodges-Lehmann Estimate	95% CL for Location Parameter		Effect size
	Mean	SD	Mean	SD			Lower	Upper	
IT (s)	19.93	14.69	16.48	9.83	0.003*	1.84	0.63	3.09	-0.128
RT (s)	2.52	1.53	2.25	1.34	0.008*	0.15	0.04	0.27	0.136
RA (%)	66.7	47.2	75.2	43.2	<0.001**	-8.24e-6	-8.11e-5	-7.10e-5	-0.085

Note: * indicates a significant p-value <0.05; ** indicates a significant p-value <0.001.

5.4. Eye movement results

5.4.1. Average eye-tracking metrics

The average fixation-based and saccade-based metrics for all stimuli are listed in Table 5. In particular, the total fixation duration was considerably lower in the FI group than in the FD group ($\Delta=2.64$ s, $n=56$, $p<0.001$). Likewise, the FI group had significantly shorter average fixation duration ($\Delta=17.16$ ms, $n=56$, $p<0.001$), less fixation count ($\Delta=7.74$, $n=56$, $p<0.05$), less saccade count ($\Delta=6.07$, $n=56$, $p<0.05$), lower peak velocity of saccades ($\Delta=12.45^\circ/\text{s}$, $n=56$, $p<0.05$), and lower average saccade amplitude ($\Delta=0.5^\circ$, $n=56$, $p<0.001$) than the FD group. These results indicate that the FI group generally had a lower cognitive load than the FD group.

Regarding the eye-tracking metrics of each dashboard, the FD group's value for each indicator was typically higher than that of the FI group, as shown in Fig. 6. Compared with the FD group, the FI group had significantly shorter fixation duration at the s5 ($\Delta=2.99$ s, $n=56$, $p<0.05$), s13 ($\Delta=11.79$ s, $n=56$, $p<0.05$), and s16 ($\Delta=5.46$ s, $n=56$, $p<0.05$) dashboard. In terms of average gaze time, there were statistical differences between the two groups for s3 ($\Delta=21.54$ ms, $n=56$, $p<0.05$), s9 ($\Delta=34.25$ ms, $n=56$, $p<0.05$), s12 ($\Delta=30.32$ ms, $n=56$, $p<0.05$), s16 ($\Delta=44.90$ ms, $n=56$, $p<0.05$) dashboard. There was a statistically significant difference between the s5 ($\Delta=12.67$, $n=56$, $p<0.05$) and s13 ($\Delta=47.26$, $n=56$, $p<0.05$) dashboards in terms of the number of gaze points. The FD and FI only showed significant differences on the s7 ($\Delta=0.93$, $n=56$, $p<0.05$) and s12 ($\Delta=1.32$, $n=56$, $p<0.05$) dashboard in terms of saccade amplitude. With each stimulus, no substantial differences in the number of saccades and average peak velocity of the saccades were found by two cognitive styles.

< 0.05), s16 ($\Delta=44.90$ ms, $n=56$, $p<0.05$) dashboard. There was a statistically significant difference between the s5 ($\Delta=12.67$, $n=56$, $p<0.05$) and s13 ($\Delta=47.26$, $n=56$, $p<0.05$) dashboards in terms of the number of gaze points. The FD and FI only showed significant differences on the s7 ($\Delta=0.93$, $n=56$, $p<0.05$) and s12 ($\Delta=1.32$, $n=56$, $p<0.05$) dashboard in terms of saccade amplitude. With each stimulus, no substantial differences in the number of saccades and average peak velocity of the saccades were found by two cognitive styles.

5.4.2. Visualization of eye-tracking records

Eye movement heat maps of the FD and FI groups for each dashboard are shown in Appendix I, with examples presented in Fig. 7. Heatmaps provided an intuitive visualization of individual relative fixation during the TOI of each stimulus presentation period, because each participant's TOI differed. A heat scale was used to depict fixation duration, ranging from cold/green to hot/red. Green indicates a shorter fixation time in the location, whereas red indicates a longer fixation duration in the region. The heat map revealed that the FD group had more fixations and longer fixation durations. The subjects attempted to find out answers based on keywords and determined whether there were relevant information based on the navigation labels. The fixation locations of each

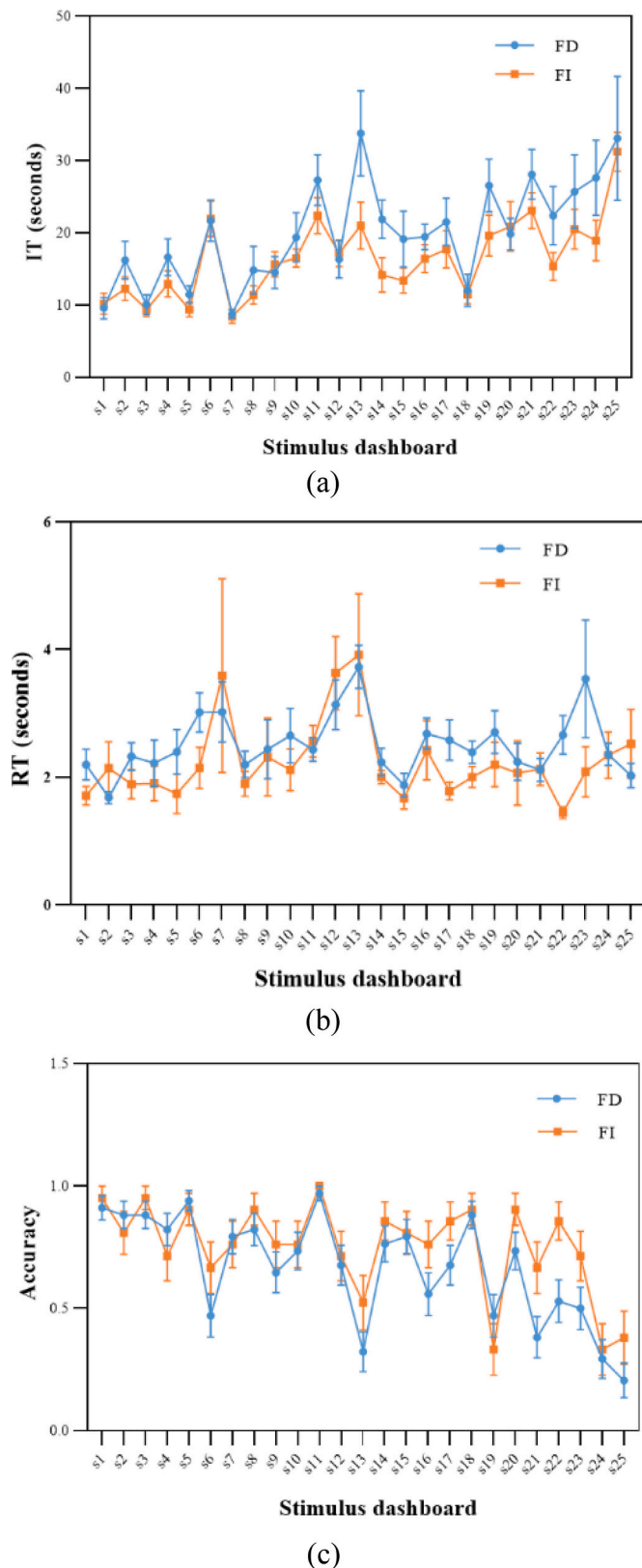


Fig. 5. Average Task performance and standard error of the mean for two groups at each dashboard composed of different levels of information load: (a) identification time; (b) response time for question; (c) response accuracy.

AOI were then scrutinized in depth to help themselves understand the fixation locations of diverse information for answers.

5.4.3. Eye-tracking metrics at AOIs

Fixation-based metrics, including the average fixation duration, number of fixations, and dwell percentages of subjects for different functional AOIs within each group, were calculated, as shown in Table 6. Regarding the average fixation duration for all AOIs, the respondents' average fixation duration for AOI was the smallest for the image and the icon and the longest for the table, while the rest of the areas fluctuate around 200 ms. The average fixation duration of the FD group was significantly longer than that of the FI group in the graph ($p < 0.05$), word ($p < 0.05$), and navigation bar areas ($p < 0.001$). Furthermore, the FD group had a considerably larger number of fixations in the data ($p < 0.05$) and navigation bar areas ($p < 0.05$) than the FI group.

The results of dwell percentages showed that participants' fixation on the graph area was the highest (FD: 38.05% vs. FI: 37.31%), followed by the word, navigation bar, and data regions. The three areas in which the participants looked the least were the icon, image, and table AOI. The specific dwell percentages of all dashboards by relevant/irrelevant AOIs are shown in Fig. 8.

The average duration of fixation in relevant and irrelevant information areas differed between the two groups (Table 7). In the relevant area, the average fixation duration in the FI group was significantly longer than that in the FD group ($p < 0.001$), whereas the average fixation duration in the FI group was substantially lower than that in the FD group in the irrelevant area ($p < 0.001$). Moreover, the number of fixations of the FI group in the irrelevant area was much lower than in the FD group ($p < 0.05$). The dwell percentages of the subjects for relevant and irrelevant AOIs within each group were calculated, as shown in Fig. 9. The two groups spent twice as much time in the irrelevant area (70.1%) as they did in the relevant area (29.2%). FD (70.5% vs. 28.6%, $F_{2,449} = 17.17$, $p < 0.001$) and FI (69.1% vs. 30.9%, $F_{2,174} = 10.34$, $p < 0.001$) showed significant differences.

To evaluate the negative impact of irrelevant information on the duration of locating relevant information, the statistics for the duration of the first fixation on relevant and irrelevant information are shown in Fig. 10. For the FD group, there was a considerable difference between relevant (4.46 ± 2.22 s) and irrelevant information (0.39 ± 1.14 s) in the time to first fixation ($F_{2,350} = 32.68$, $p < 0.001$). There was a significant difference ($F_{2,350} = 20.38$, $p < 0.001$) in the time to the first fixation between relevant (3.97 ± 2.19 s) and irrelevant (0.26 ± 0.66 s) information for the FI group.

5.4.4. Correlation and regression results of information load

The correlation between cognitive load and information load measured by the eye movement index is shown in Table 8 to explore the link between cognitive load and information load. Three of the six indicators in both groups were shown to be statistically related to information load; therefore, the total fixation duration, fixation count, and saccade count were further utilized for regression analysis.

Fig. 11 shows the regression curves of the information load on cognitive load based on two cognitive styles. Their coefficient results are following presented in Table 9. Despite the relatively low adjusted R^2 values indicating that the regression analysis results are not statistically meaningful for prediction, the relationship between information load and cognitive load might be interpreted for reference. Two substantial knots in the connection were discovered at roughly 0 and 10,000 based on three main eye tracking metrics. After removing outliers detected by the box plot, natural spline regression was used with two knots and piecewise functions to match the association between information load and cognitive load.

Through preliminary fitting, the two nodes were set to 0 and 10,000 respectively. Although three models of total fixation duration ($F_{3,693} = 21.25$, $p < 0.001$), fixation count ($F_{3,698} = 23.07$, $p < 0.001$), and saccade count ($F_{3,701} = 19.1$, $p < 0.001$) were significant, they did not fit

Table 5

Eye-tracking metrics comparison of the FD and FI group.

Indicators	Mean $\pm$ SD		p-value	Hodgea-Lehmann Estimate	95% CI for Hodgea-Lehmann Estimate		Effect size Rank-Biserial Correlation
	FD	FI			Lower	Upper	
Total fixation duration (s)	12.89 $\pm$ 10.49	10.25 $\pm$ 7.35	<0.001**	1.79	0.90	2.70	0.17
Average duration (ms)	213.38 $\pm$ 42.79	196.22 $\pm$ 41.06	<0.001**	18.00	12.00	23.00	0.26
Fixation count	58.84 $\pm$ 44.80	51.10 $\pm$ 31.79	0.022*	5.00	1.00	5.00	0.10
Saccade count	44.89 $\pm$ 35.44	38.82 $\pm$ 27.36	0.008*	5.00	1.00	8.00	0.16
Average peak velocity of saccades ($^{\circ}$ /s)	191.44 $\pm$ 46.12	178.99 $\pm$ 53.48	0.004*	10.46	3.33	17.79	0.13
Average amplitude of saccades ($^{\circ}$)	5.10 $\pm$ 1.40	4.60 $\pm$ 1.56	<0.001**	0.49	0.29	0.70	0.20

Note: * indicates a significant p-value <0.05; ** indicates a significant p-value <0.001.

the original data well, according to the adjusted R^2 (0.08, 0.09, and 0.07). In the three models, the estimated coefficients of the last two splines are significant.

Multilevel linear regression analyses were performed with independent variables (information characteristics, task and process parameters, quantity of information, and cognitive styles) to explain their correlations with cognitive load assessed by the whole fixation duration, fixation count, and saccade count. To better explain all independent variables' fixed effects on cognitive load, individual differences were taken into account as a random effect. The group of cognitive styles used dummy variables with values of 0 and 1. The unit of fixation time was adjusted in seconds to match the scale with the independent variable. Table 10 summarizes the results of the multivariate linear regression after removing outliers (total fixation duration exceeding 40s, fixation count, and saccade count surpassing 200). All regression models indicated statistically significant results for whole fixation duration ($F_{4,692} = 20.93$, $p < 0.001$), fixation count ($F_{4,697} = 20.07$, $p < 0.001$), and saccade count ($F_{4,700} = 17.09$, $p < 0.001$). Information characteristics had a significant negative influence on total fixation duration; on the other hand, the quantity of information and cognitive style were significantly correlated with total fixation duration. The impact of information characteristics on fixation counts was significantly negative ($p = 0.004$), whereas that of quantity was significantly positive ($p = 0.004$). Saccade count was negatively correlated with information features ($p = 0.04$), while it was positively correlated with task process factors, quantity ($p = 0.001$), and cognitive style. Three regression models revealed that all variables explained approximately 30% of the variance in the cognitive load. Information characteristics have a substantial impact on cognitive load; especially when the information characteristic increases by one, the cognitive load evaluated by fixation counts decreases by 4.64, controlling for other parameters. The models for fixation duration and saccade count had better R^2 values than those for fixation count.

6. Discussion

6.1. Visual attention for dashboard

The area with the most dwell proportion and the area with the least dwell proportion did not vary significantly as the information load increased. In the process of identifying information in the dashboard, participants typically linger in graphic areas, followed by the words and navigation area. Attention at AOs does not preclude the guidance of questioning, and free gaze and selection may yield different results. Professional monitoring with dashboards is usually task-oriented and involves visual search rather than unconscious browsing and visual exploration [95]. The average fixation period in the three most concentrated regions was approximately 200 ms, indicating that the participants considered the graph, words, and navigation areas to be the most critical, and that they carefully digested the information in these three areas. Graphics were the most engaging and contained the most intuitive features, validating the figure-ground and closure principles of

classic Gestalt theory [96].

The least attention was paid to images, icons, and tables; however, the average time spent gazing at these three components differed slightly. Not only did the percentage of images and icons remain low, but the short duration of each gaze suggested that the respondents perceived these two types of information as insufficient or unimportant. The fact that images and icons do not convey specific information and signs in this study, but rather serve as schematics, is one of the reasons why participants paid less attention to them. Although the table had the least dwell attention of the participants, the average fixation duration was the longest, suggesting that organizing information in tables increases the difficulty of comprehension on the dashboard. There are contradictory findings of graphical and tabular representations, one of which found that when decision-makers had a limited understanding of accounting, using customer profitability statistics given in a graphical format rather than a tabular format, they made decisions that resulted in higher profits [97]. However, when the complexity of the tasks increased, the tabular representations proved to be superior to the graphical format [98]. The use of charts or tables in data panels depends on the nature of the task. The tables are more suitable for extracting symbolic information for precise values, whereas graphs and tables are more suitable for displaying spatial information to understand relationships and comparisons [54].

Irrelevant information distracted the participants and the fixation rate in the irrelevant area was almost twice as high as that in the relevant region. As the information load increased, there was little change in the percentage of dwellings in the relevant information area compared to that in the irrelevant information area. In the early stages of the increase in information load, participants spent less time locating relevant information, which may have been a learning and adaptation process. However, as the quantity of information presented increased, the time to the first fixation on relevant information also increased, with no discernible improvement as the amount of information offered increased.

6.2. The role of cognitive style in cognitive load

The H2 hypothesis was confirmed for both task performance and cognitive load, as measured by eye movement metrics. Cognitive style was significantly correlated with the visual search time and response accuracy. The FI group required less time to identify information and achieved a higher accuracy. This can be explained by their more oriented and organized visual search strategies [99].

Cognitive load was significantly higher in the FD group than in the FI group, reflected by greater fixation duration and number of fixations and saccades, which may be due to the longer average gaze duration and number of gazes for irrelevant information in the FD group. The same finding revealed that the FD group exhibited more fixations and saccades than the FI group, consistent with the patterns in previous studies [99]. However, some study results conflict with this finding; the FI group had a longer fixation duration [78]. A study of consumer behavior found that FI customers had longer fixation durations and spent more time

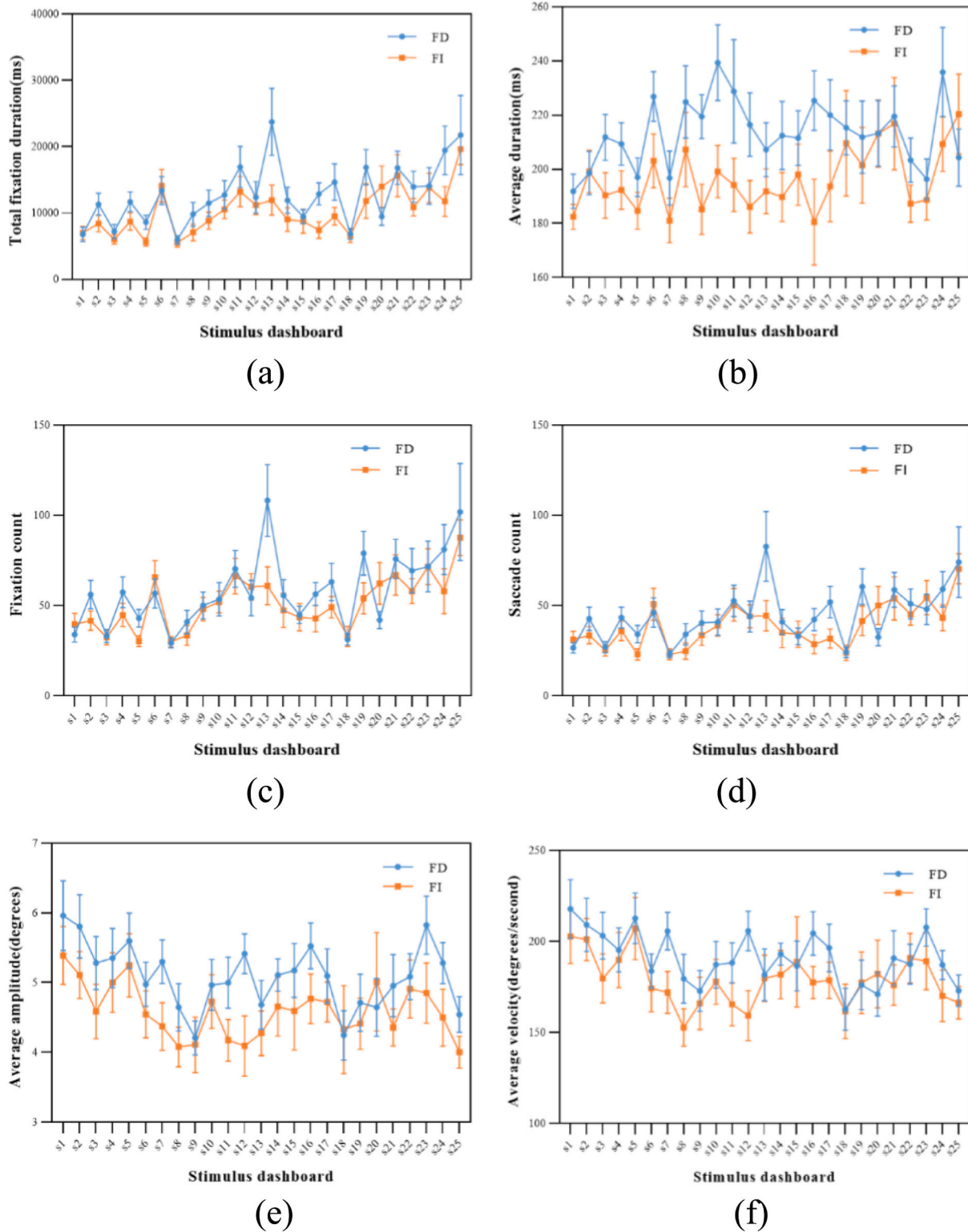


Fig. 6. Group-level average eye movement metrics occurred within each TOI: (a) represents total fixation duration; (b) represents average fixation duration; (c) represents fixation count; (d) represents saccade count; (e) represents the average amplitude of saccades; (f) represents the average velocity of saccades.

assessing product labels, as well as a stronger tendency to seek information more exhaustively [78]. The same results were found in a study on interactive game elements, fixation count, and duration were higher in FI players than in FD gamers. The FD players were less likely to notice details, but FI players were more analytical and spent more fixations [77]. The differences in eye movement measures between the two

groups did not widen as the information load increased and even tended to cross over when the information load reached its peak (Fig. 11). This finding contradicts previous findings that the FI group is more concerned with objective and complex information, being more adaptive and efficient when the perceptual load is sharp in the field of view of changing roads [100,101]. Although there was no significant difference



(a)



(b)

Fig. 7. A heat map example of average relative fixation duration between (a) FI and (b) FD group.

Table 6

Average fixation-based metrics at functional AOIs for the FD and FI group.

Area of Interest	Parameter	Mean $\pm$ SD		p-value
		FD	FI	
Data	Average fixation duration (s)	0.21 $\pm$ 0.09	0.20 $\pm$ 0.08	0.40
	Fixation count	4.15 $\pm$ 5.75	3.63 $\pm$ 5.61	0.04*
	Dwell percentage (%)	12.7 $\pm$ 13.2	11.7 $\pm$ 11.3	0.94
	Average fixation duration (s)	0.21 $\pm$ 0.09	0.19 $\pm$ 0.07	0.03*
Graph	Fixation count	15.11 $\pm$ 22.52	12.21 $\pm$ 14.76	0.25
	Dwell percentage (%)	38.05 $\pm$ 27.3	37.3 $\pm$ 26.2	0.95
	Average fixation duration (s)	0.18 $\pm$ 0.08	0.17 $\pm$ 0.05	0.08
	Fixation count	1.75 $\pm$ 3.92	1.36 $\pm$ 2.99	0.19
Icon	Dwell percentage (%)	3.8 $\pm$ 7.9	4.2 $\pm$ 8.0	0.86
	Average fixation duration (s)	0.17 $\pm$ 0.06	0.17 $\pm$ 0.07	0.66
	Fixation count	1.90 $\pm$ 4.58	1.35 $\pm$ 3.34	0.21
	Dwell percentage (%)	3.6 $\pm$ 5.8	3.9 $\pm$ 5.1	0.77
Image	Average fixation duration (s)	0.21 $\pm$ 0.08	0.19 $\pm$ 0.08	<0.001**
	Fixation count	5.57 $\pm$ 8.51	4.34 $\pm$ 6.11	0.003*
	Dwell percentage (%)	13.0 $\pm$ 7.8	14.2 $\pm$ 7.8	0.69
	Average fixation duration (s)	0.22 $\pm$ 0.07	0.20 $\pm$ 0.07	0.01*
Navigation bar	Fixation count	9.50 $\pm$ 12.29	8.50 $\pm$ 10.80	0.21
	Dwell percentage (%)	27.43 $\pm$ 21.7	26.86 $\pm$ 21.3	0.88
	Average fixation duration (s)	0.26 $\pm$ 0.08	0.23 $\pm$ 0.09	0.21
	Fixation count	0.59 $\pm$ 3.22	0.29 $\pm$ 1.88	0.69
Word	Dwell percentage (%)	1.3 $\pm$ 6.5	1.9 $\pm$ 9.5	0.90
	Average fixation duration (s)	0.26 $\pm$ 0.08	0.23 $\pm$ 0.09	0.21
	Fixation count	0.59 $\pm$ 3.22	0.29 $\pm$ 1.88	0.69
	Dwell percentage (%)	1.3 $\pm$ 6.5	1.9 $\pm$ 9.5	0.90

Note: * indicates a significant p-value <0.05; ** indicates a significant p-value <0.001.

between the two groups in the proportion of attention allocated to functional regions, the FD group had a higher cognitive load (longer average gaze duration) in critical and most focused areas, such as graphs, words, and navigation bars. The finding is consistent with the first study aimed at investigating the effect of FD/FI style on

instructional Web interactions through eye-tracking [102].

The longer average fixation duration in the relevant information regions of FI individuals facilitated the acquisition and comprehension of critical information. By contrast, the longer average fixation duration and a large number of fixations located in irrelevant information regions of FD individuals contributed to the interference of irrelevant information. Previous research has indicated that FI individuals can focus and maintain more attention on relevant information while suppressing their attention to irrelevant information than FD individuals [103]. Another study claimed that FIs could filter out distractors more effectively than FDs from cognitive and neural mechanisms [104]. However, in this study, the average fixation duration rather than the dwell percentage reflected the inhibition of attention to irrelevant information.

6.3. Relationship between information load and cognitive load

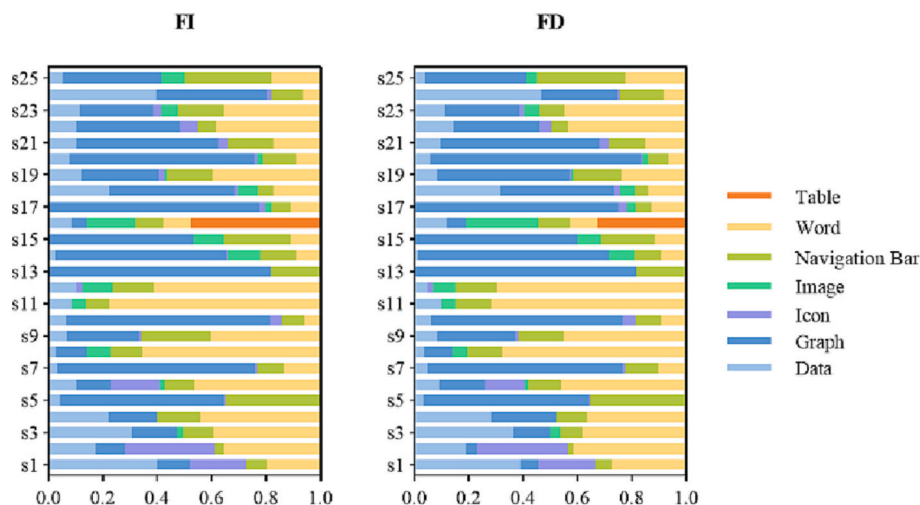
The trend of cognitive load with varying information load failed to confirm the H1 hypothesis. This study is characterized by three separate correlations, indicating two major knots in the entire relationship. One node emerges in relation to the learning and adaptation processes, whereas the other node emerges as a cutoff point beyond cognitive resources. This finding contradicts previous evidence of a positive linear or inverted U-shaped relationship between information presentation and other cognitive-related indicators such as working memory [105,106]. The three multilevel regression models indicated that the effect of

Table 7

Average fixation-based metrics at relevant and irrelevant AOIs for the FD and FI group.

Area of Interest	Parameter	Mean $\pm$ SD		p-value
		FD	FI	
Relevant	Average fixation duration (s)	0.23 $\pm$ 0.09	0.26 $\pm$ 0.10	<0.001**
	Fixation count	11.99 $\pm$ 10.78	11.04 $\pm$ 10.81	0.17
	Dwell percentage (%)	28.6 $\pm$ 26.0	30.9 $\pm$ 24.5	0.17
	Average fixation duration (s)	0.19 $\pm$ 0.04	0.18 $\pm$ 0.04	<0.001**
Irrelevant	Fixation count	38.99 $\pm$ 41.84	30.06 $\pm$ 26.09	0.002*
	Dwell percentage (%)	70.5 $\pm$ 26.7	69.1 $\pm$ 24.5	0.30

Note: * indicates a significant p-value <0.05; ** indicates a significant p-value <0.001.

**Fig. 8.** Dwell percentages at the functional AOI of each dashboard.

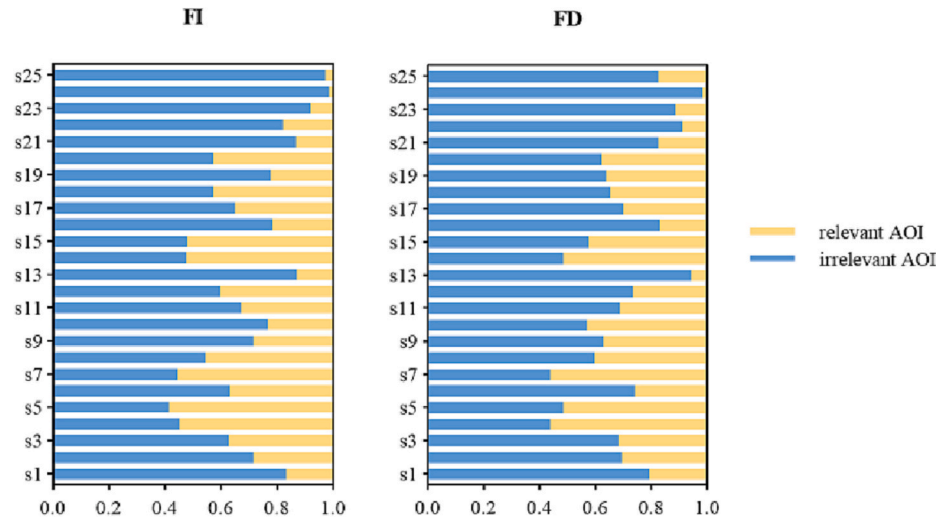


Fig. 9. Dwell percentages at relevant and irrelevant AOI.

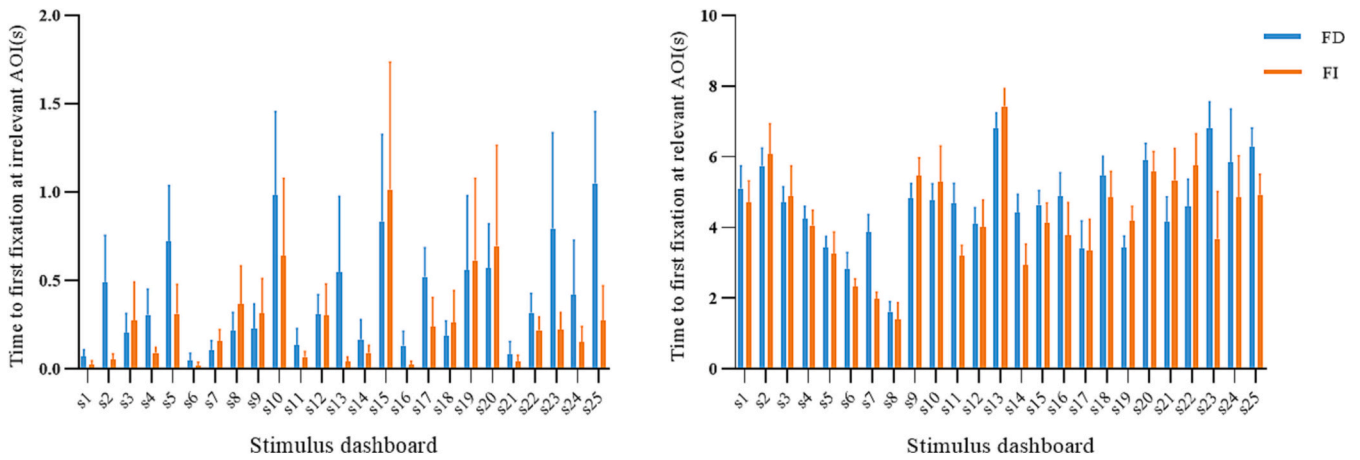


Fig. 10. Time to the first fixation at relevant and irrelevant AOIs.

Table 8
Correlation between eye-tracking metrics and information load.

Indicators	Group	Pearson r	95% confidence interval		R squared	p-value (two- tailed)
			Lower	Upper		
Total fixation duration	FD	0.70	0.42	0.86	0.49	<0.001**
	FI	0.63	0.31	0.82	0.39	0.001*
Average duration	FD	0.49	0.12	0.74	0.24	0.013*
	FI	0.16	-0.25	0.53	0.03	0.434
Fixation count	FD	0.70	0.42	0.86	0.49	<0.001**
	FI	0.63	0.32	0.82	0.40	0.001*
Saccade count	FD	0.67	0.37	0.84	0.44	<0.001**
	FI	0.58	0.24	0.79	0.34	0.02*
Average peak velocity of saccades	FD	-0.43	-0.71	-0.04	0.19	0.031*
	FI	-0.29	-0.62	-0.11	0.09	0.154
Average amplitude of saccades	FD	-0.29	-0.62	0.12	0.08	0.158
	FI	-0.32	-0.64	0.08	0.10	0.117

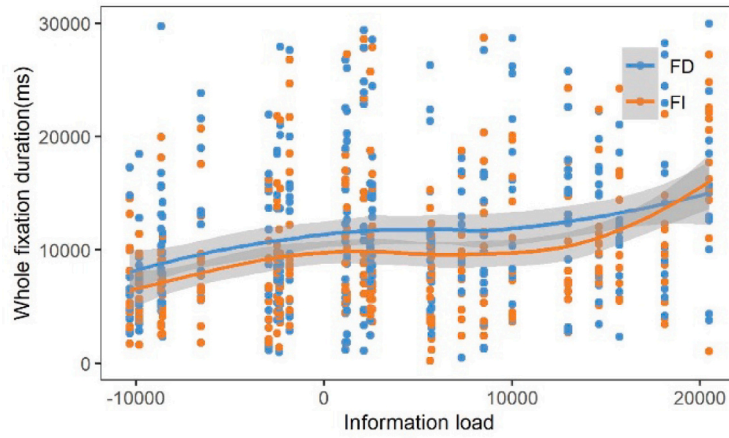
Note: * indicates a significant p-value <0.05; ** indicates a significant p-value <0.001.

information load on cognition was significant, with the cognitive load increasing as the complexity, ambiguity, unclarity, and novelty of information increased. In contrast, task and process parameters, which

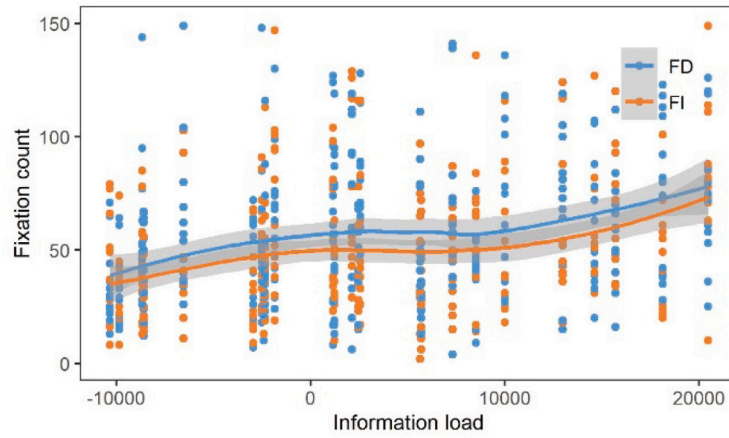
were primarily moderated by the number of dashboard modules, had no significant effect on cognitive load, except for the saccade count. The models for fixation duration and saccade count had better R^2 values than the models for fixation count in terms of regression model goodness-of-fit, demonstrating that fixation duration and saccade count may be more valuable than fixation count in terms of measuring cognitive load.

6.4. Practical implications

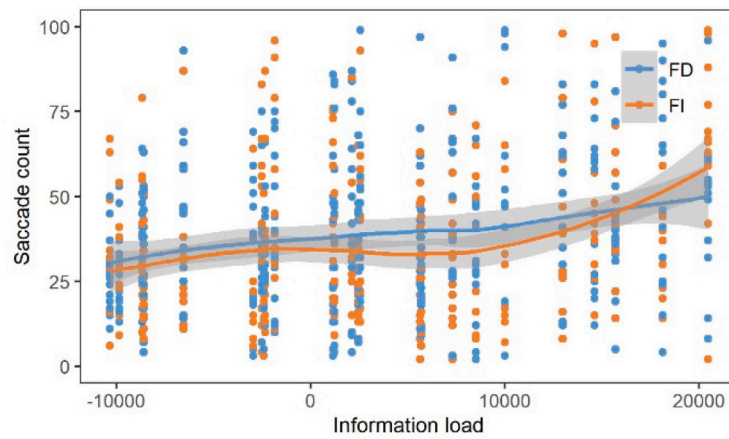
Understanding how dashboard information load affects cognitive load makes great contributions to construction practice. Firstly, findings in this research can provide a theory-driven guideline for designing cognitive-fit dashboards for project managers or other relevant construction professionals. The design for dashboards usually includes two aspects: visual features and functional features. From a visual perspective, structured modules and data aggregation on dashboards are suggested to be created with caution when it comes to information quantity, since those with nine or more project-related information modules may easily overburden users. Dashboards with four or six modules can be prioritized, which allow individuals to have enough cognitive resources for information processing. Concerning human's limited span of visual attention, sound navigation [7] might be integrated for users when they use dashboards with too many dimensions of project information. In terms of display formats designed to serve as decision cues on dashboards, usage of graphs by default appears to be justified, given that they



(a)



(b)



(c)

Fig. 11. FD and FI spline regression results despite the adjusted $R^2 < 0.1$. (a). total fixation duration regression; (b) fixation count regression; (c). saccade count regression.

Table 9

Natural spline regression between information load on cognitive load variables.

Dependent variables	Piecewise	Estimate	SE	t	p-value	Adjusted R ²
Total fixation duration					3.56e-13***	0.08
	Intercept	7216.8	698.4	10.33	2e-16***	
	Spl 1	2211.4	1198.6	1.85	0.066	
	Spl 2	11,060.2	1798.4	6.18	1.09e-9***	
Fixation count	Spl 3	6247.4	1007.0	6.20	9.48e-10***	0.09
	Intercept	36.81	3.10	11.87	3.01e-14***	
	Spl 1	9.89	5.30	1.87	2e-16***	
	Spl 2	49.05	7.92	6.19	0.062	
Saccade count	Spl 3	29.77	4.42	6.74	1.01e-9***	0.07
	Intercept	29.45	2.71	10.86	3.36e-11***	
	Spl 1	7.37	4.63	1.59	6.44e-12	
	Spl 2	35.59	6.92	5.14	2e-16***	
	Spl 3	25.03	3.84	6.52	0.112	
					3.55e-7***	
					1.35e-10***	

Note: SE means standard error. * means $p < 0.05$, ** means $p < 0.01$, *** means $p < 0.001$.

are more likely to be preferred based on eye-tracking measurements. More options for switching to another display format, such as a textual description of the project, might be supplied as an alternative, if required. However, dashboard designers should avoid using flashy images, which were found to be the least appealing in this experiment. Instead, they can be autonomously monitored by Artificial Intelligence, which can transform them into smart data for computer-assisted decision making on construction sites. Furthermore, the questioning guide may help project managers operate dashboards more easily, and question settings are preferred by some users in the construction area mentioned in previous study [107]. From a functional standpoint, an interface with a cognitive style test might be introduced before project managers access the dashboard display. This could therefore enable run-time customization of user interfaces and cognitive-fit presentation

formats to cater two user types with varied cognitive styles. Furthermore, because FD individuals are more likely to be distracted by irrelevant information and produce more cognitive load than FI individuals, a simpler interface with main navigation might be needed. In this setting, a point-and-click interaction could be designed specifically for the FD user interface, allowing them to freely select any project information they desire by digging down the primary navigation. In general, construction dashboard designers should optimize distinct visual and functional perceptions to create a more clear, concise, and intuitive interface to guide rather than distracting users' attention in smart construction management according to two different cognitive styles. Secondly, it has practical implications for task-orientated staffing in construction. During the pre-service screening, based on the identification of individual cognitive styles, it helps choose the most suitable people with a matching cognitive style to be managers or specialists in varied working scenarios. For example, site managers with the FI style rather than the FD style may be more competent in smart management of mega construction projects, in which the dashboard with a relatively high level of information load may be more likely to be used. Prior to taking the role, the feedback from measuring eye movements that illustrates the influence on cognitive load can make them aware of their strengths and shortcomings in dashboard use and attention distribution. Furthermore, because directed attention allocation is a skill that can be learned and improved, potential users whose eye-tracking indicators reveal a mismatch to dashboard information requirements could be provided with procedural- or attentional-based training to remedy the dysfunctional effect caused by massive information load. Thirdly, this study offers a starting point for customized information visualization user interfaces in the construction management paradigm shift toward digitalization and automation. Although creating a customized dashboard for each user is utopic, designing a dashboard based on distinct cognitive styles could become an efficient customization approach for information visualization in the future construction sector.

6.5. Limitations and prospects

This study has a few limitations that require further research. First, although multiple levels of information load were intentionally established in this work to investigate its impact on cognitive load, their relationships may be influenced by variances in experimental designs, such as contextual range restriction, underestimation of the underload condition, independent variable numbers, or measures of performance [108]. In future designs, it is expected to perform repeated measurements for each participant on each level with considering more variables to achieve higher accuracy. Besides, the validation of dashboard

Table 10

Linear mixed model regression for effects of independent factors on cognitive load variables.

Dependent variables	Independent factors	Estimate	SE	t	p-value	Adjusted R ²
Total fixation duration	Intercept	11.80	4.04	2.92	<0.001***	0.30
	IC	-1.11	0.36	-3.13	0.004**	
	TPP	1.26	0.69	-1.82	0.002**	
	Quant	0.21	0.08	2.57	0.07	
	Group	1.77	0.52	3.39	0.01*	
	Fit of model				0.001**	
Fixation count	Intercept	53.05	18.04	2.94	<0.001***	0.24
	IC	-4.64	1.59	-2.92	0.003**	
	TPP	6.09	3.08	-1.98	0.004**	
	Quant	1.02	0.36	2.87	0.05	
	Group	4.26	2.33	1.83	0.004**	
	Fit of model				0.07	
Saccade count	Intercept	29.97	15.72	1.91	<0.001***	0.30
	IC	-2.91	1.39	-2.09	0.06	
	TPP	6.08	2.69	-2.26	0.04*	
	Quant	1.01	0.31	3.25	0.02*	
	Group	4.37	2.03	2.14	0.001**	
	Fit of model				0.03*	

Note: SE means standard error. * means $p < 0.05$, ** means $p < 0.01$, *** means $p < 0.001$.

information load calculation model needs to be further explored in more ways, despite the fact that it is adequately theory-based developed. Future research might focus on establishing quantitative criteria for computing the information load of visualized dashboards. Second, the participants in this study were students, leading to their characteristics in greater homogeneity within the group. As a result, the values of some parameters (e.g., those related to personal factors) were assumed to be the same during the calculating process of dashboard information load. Professional monitoring personnel with the actual working experience will be invited for experiment engagement in the future work. Meanwhile, concerning the widespread use of graphical dashboards and the growing number of users, the study of cognitive load could be extended to encompass users with distinct individual characteristics, such as those of various educational levels and professional backgrounds. Third, although eye tracking is a popular technology for measuring cognitive-related variables in HCI research, cognitive processing and information interpretation still cannot be directly inferred from eye tracking results. More physiological measurement tools, such as EEG, could be added for quantifying how different persons process and interpret dashboard information differently when the dashboard information load varies. Additionally, calibrating participants with droopy eyelids, hard contact lenses, narrow glasses, or who are wearing mascara during the calibration procedure prior to collecting eye tracking data may be difficult or impossible, resulting in some low-quality sample data that had to be discarded. Despite these limitations, findings of this research still contribute to understanding how information load of construction dashboards and users' cognitive styles affect cognitive load from a theoretical perspective of the individual cognitive mechanism based on eye-tracking technologies, and have practical implications for HCI advancement in the construction industry.

7. Conclusion

This study provides preliminary insights into the effects of information load and cognitive style on cognitive load of visualized dashboard used for construction-related activities through an eye movement visual search experiment. Individuals were found to be overwhelmed with dashboards having nine or more information modules. FD participants took longer to access information on dashboards and had lower response accuracy than FI participants. Cognitive load was significantly higher in the FD group than in the FI group as measured by fixation duration, fixation count, and saccade count. The average fixation duration of FI participants was longer in relevant areas, while FD participants fixated for longer in irrelevant areas, as well as more fixations. The relationship between information load and cognitive load suggests that individual nodes may result in segmental linear relationships with varying slopes, not totally positive associations or inverted U-shaped curves.

The novelty of this study can be found on both theoretical and practical levels. For academics, this study is one of the first attempts in the field of construction management to measure the relationship between dashboard information load on users' cognitive load while taking cognitive styles into account. Such an effort can increase a fundamental understanding of the role of information features and individual traits in user cognitive efficiency when confronted with construction-related HCI. Moreover, the results of this study validate the utility of using objective eye-tracking metrics to fairly test how information load affects cognitive load in two types of users varying in cognitive styles in contrast to earlier studies using self-reported investigation. Therefore, this approach can be broadened to conduct cognitive science research on human-oriented construction management.

For construction practices, this study firstly provides a new view for cognitive-fit dashboard design in the construction industry compared with prior research focusing on project-centered design. Based on the findings of this study, dashboard designers will be better equipped to deliver effective information on dashboards that are suitable for diverse

user types. Second, it contributes to task-oriented staffing and attention-based training for project managers during pre-service screening, allowing them to better match the information requirement in a specific working scenario through improved cognitive methods. Furthermore, by demonstrating the practical feasibility of cognitive style-based dashboards in smart construction management, this study provides a starting point for the creation of customized information visualization user interfaces in Construction 5.0. Further work may look into the potential of other individual cognitive or contextual factors influencing cognitive load and information processing efficiency, which will help guide the development of customized visualization systems while also facilitating the transition from the physical to the digital world in the construction industry.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.autcon.2023.105029>.

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